STAT 5242/4242 HW3 — Problem 4: Solutions

(a) Backpropagation through one layer

The Jacobian is $D_\ell W_\ell$. For column gradients,

$$g_{\ell-1} = W_\ell^T D_\ell g_\ell, \qquad g_0 = W_1^T D_1 W_2^T D_2 \cdots W_L^T D_L g_L.$$

(b) Sigmoid and vanishing gradients

For sigmoid, $\phi'(t) = \phi(t)(1-\phi(t)) \in [0, 1/4]$, so $\|D_\ell\|_2 \le 1/4$.

$$\|g_{\ell-1}\|_2 \le \|W_\ell\|_2 \|D_\ell\|_2 \|g_\ell\|_2 \le (s/4)\|g_\ell\|_2.$$

Iterating, and using $\|g_L\|_2 > 0$, gives $\|g_0\|_2 / \|g_L\|_2 \le (s/4)^L$.

For 0 < s < 4, the bound guarantees exponential decay. For s > 4,

a growing upper bound does not prove growth: the actual ratio may vanish.

(c) ReLU

All preactivations are positive, so $D_\ell = I_d$ and $g_0 = a^L g_L$. Thus

$\|g_0\|_2 / \|g_L\|_2 = a^L$. This tends to zero for $0 < a < 1$, equals one for

a = 1, and grows without bound for a > 1. ReLU alone does not guarantee

stable propagation: the weight scale matters.

(d) Scaled residual connections

Let $B_\ell = I_d + A_\ell/L$. Backpropagation gives $g_{\ell-1} = B_\ell^T g_\ell$.

The triangle and reverse triangle inequalities give, for each layer,

$$(1 - C/L)\|g_\ell\|_2 \le \|g_{\ell-1}\|_2 \le (1 + C/L)\|g_\ell\|_2.$$

The lower factor is nonnegative because $L \ge 2C$. Iterating yields

$$(1 - C/L)^L \le \|g_0\|_2/\|g_L\|_2 \le (1 + C/L)^L.$$

Using the hint with $u = C/L \le 1/2$, and raising to the Lth power,

$$e^{-2C} \le \|g_0\|_2/\|g_L\|_2 \le e^C.$$

Both bounds are positive, finite and independent of depth. Thus the ratio

cannot tend to zero or infinity solely as depth increases under these assumptions.